

Education

University of California, Berkeley PhD in Electrical Engineering and Computer Sciences	Fall 2021 - Spring 2026
University of California, Berkeley B.S. in Electrical Engineering and Computer Sciences Graduated with Highest Honors GPA: 4.0 / 4.0	Fall 2017 - Spring 2021

Research Interests

Reinforcement learning; safe learning in control; robust control; control theory and optimization.
Application of machine learning to control, with an emphasis on safety, particularly robustness to model uncertainty and disturbances.

Technical Skills

Methods: Machine learning, robust & nonlinear control, optimal control, optimization.
Languages: Python, C++, MATLAB, Rust, C.
Software & Tools: PyTorch, Linux, Simulink, Git, CVX

Experience

Graduate Student Researcher, UC Berkeley	Fall 2021 - Spring 2026
• Developed reinforcement learning framework to train neural network controllers to maximize reward subject to closed-loop stability and robustness requirements, aimed at improving performance in systems where reliability is critical. • Used adversarial training to train neural network controllers to maximize local robust stability, verifying properties post-training using branch-and-bound methods. • Improved accuracy of learned models of systems and sample efficiency of training process by leveraging model symmetries. • Accelerated trajectory generation in asset-guarding games through sequential convex programming for optimal control and learning-based solutions of assignment problems. • Reduced conservativeness in verification of nonlinear and uncertain systems by developing optimization-based methods for tighter characterization of nonlinearities, focused on analyzing stability and robustness. Combined with sum-of-squares programming to enlarge verified region of attraction by as much as 16 times.	
ADAS Controller Calibration Intern at General Motors	Summer 2025
• Accelerated controller development by implementing an offline reinforcement learning pipeline to streamline manual tuning; leveraged pre-existing datasets to produce controllers meeting performance benchmarks in simulation.	
Power System Control and Optimization Intern at Siemens	Spring 2023
• Improved scalability of contingency management by designing an algorithm to group contingencies (power line failures) by similarity and synthesize one robust controller per group. • Work published in IEEE Transactions on Power Systems.	

Embedded Software Simulation Intern at Waymo

Summer 2020

- Designed and implemented a C++ simulation framework using the Functional Mock-up Interface (FMI) standard, enabling simulation of interconnected vehicle subsystem models, reducing model integration overhead for simulation team.

Avionics Lead, Space Technologies and Rocketry Club

2019 - 2021

- Directed design and implementation of flight computer and telemetry systems across all stages: hardware, firmware and software, and integration with other rocketry subsystems.
- Designed an ARM microcontroller-based ground station (schematic, PCB layout, and firmware) for receiving telemetry.

Autoflight Intern for 777X Program at Boeing

Summer 2019

- Created monitors of autonomous takeoff. Modeling in Simulink, simulation of models, and analysis of data with MATLAB.
- Software integration support (C and C++) for various projects, using the ARINC 429 protocol to interface between different systems.

Intelligent Robotics Group Intern at NASA Ames

Summer 2018

- Research and implementation of algorithms for relative localization of nodes in a wireless sensor network using ultra-wide band radios for pair-wise distance measurement.
- Hardware and firmware prototyping of nodes.

Undergraduate researcher in UC Berkeley Swarm lab (Advisor: Kristofer Pister)

Spring 2018 - Spring 2019

- Implemented and compared computationally-inexpensive pose-estimation algorithms fusing IMU data on ARM Cortex-M4 processors.

Works Under Review

- **Neelay Junnarkar**, Yasin Sonmez, and Murat Arcak. *Learning Neural Network Controllers with Certified Robust Performance via Adversarial Training*. 2026. arXiv: 2604.01188 [eess.SY]. URL: <https://arxiv.org/abs/2604.01188>

Publications

- **Neelay Junnarkar**, Peter Seiler, and Murat Arcak. “Polynomial Constraints for Robustness Analysis of Nonlinear Systems”. In: *IEEE Control Systems Letters* (2026), pp. 1–1. DOI: 10.1109/LCSYS.2026.3701072
- **Neelay Junnarkar**, Murat Arcak, and Peter Seiler. *Synthesizing Neural Network Controllers with Closed-Loop Dissipativity Guarantees*. Accepted to IFAC Automatica. 2024. arXiv: 2404.07373 [eess.SY]. URL: <https://arxiv.org/pdf/2404.07373>
- **Neelay Junnarkar**, Can Kızılkale, Nevena Golubovic, Murat Arcak, and Aydın Buluç. “Sempervirens: A Fast Reconstruction Algorithm for Noisy and Incomplete Binary Matrix Representations of Trees”. In: *INFORMS Journal on Computing* (2025). DOI: 10.1287/ijoc.2023.0373
- **Neelay Junnarkar**, Murat Arcak, and Peter Seiler. “Stability Margins of Neural Network Controllers”. In: *2025 American Control Conference (ACC)*. 2025, pp. 1355–1360. DOI: 10.23919/ACC63710.2025.11107746. URL: <https://arxiv.org/pdf/2409.09184>

- Emily Jensen, **Neelay Junnarkar**, Murat Arcak, Xiaofan Wu, and Suat Gumussoy. “Certifying Stability and Performance of Uncertain Differential-Algebraic Systems: A Dissipativity Framework”. In: *IEEE Transactions on Control of Network Systems* (2024). DOI: 10.1109/TCNS.2024.3510585
- Yasin Sonmez, **Neelay Junnarkar**, and Murat Arcak. “Exploiting Symmetry in Dynamics for Model-Based Reinforcement Learning With Asymmetric Rewards”. In: *IEEE Control Systems Letters* 8 (2024). DOI: 10.1109/LCSYS.2024.3409560. URL: <https://arxiv.org/pdf/2403.19024>
- **Neelay Junnarkar**, Emily Jensen, Xiaofan Wu, Suat Gumussoy, and Murat Arcak. “Grouping of N-1 Contingencies for Controller Synthesis: A Study for Power Line Failures”. In: *IEEE Transactions on Power Systems* (2024). DOI: 10.1109/TPWRS.2024.3393866. URL: <https://arxiv.org/pdf/2404.07415>
- **Neelay Junnarkar**, He Yin, Fangda Gu, Murat Arcak, and Peter Seiler. “Synthesis of Stabilizing Recurrent Equilibrium Network Controllers”. In: *2022 IEEE 61st Conference on Decision and Control (CDC)*. 2022. DOI: 10.1109/CDC51059.2022.9992684
- **Neelay Junnarkar**, Emmanuel Sin, Peter Seiler, Douglas Philbrick, and Murat Arcak. “Fast Assignment in Asset-Guarding Engagements using Function Approximation”. In: *2022 American Control Conference (ACC)*. 2022. DOI: 10.23919/ACC53348.2022.9867720

Teaching Experience

Teaching Assistant for Linear System Theory Fall 2022, Fall 2025

Taught recitation section, developed homeworks, graded homeworks and exams.

Teaching Assistant for Convex Optimization Spring 2026

Led discussion sections, developed and graded homeworks and exams.